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CryptoForget: A Cryptographically Verifiable Machine Unlearning Framework for Production MLOps Systems

Abstract

The personal information of people, such as financial transactions, medical history, academic performance, and behavioral data is used to train the ML processes of today. Some laws, including the GDPR Article 17 and CCPA, dictate that if a user requests to delete their personal data, organizations must comply. However, while this means that an organization must remove the personal details from the database, the training data remain part of the model. CryptoForget project was born out of the need for a system that demonstrates this fact and solves the problem on a small scale. The system consists of a working ML model with an MLOps pipeline (training, monitoring, review, retraining) and a virtual population of 15-20 entities, generating data through this model. Each entity has a function of cancellation of data. The system involves cryptographic commitment to the training data during its collection, generates a verifiable signed certificate of deletion after data erasure, and provides detailed reports about the performance of the model before and after erasing a user's data.

Introduction

Today's ML solutions, be it a loan approval system, a diagnostic support application, or a personalized learning platform, have one thing in common: they learn based on the data provided by live users. When this data is processed, it becomes dissipated in the system and ceases to exist as an identifiable entry in a database. If the users demand that their data be erased, the company may comply with their request to delete the entry, but it could not guarantee that the impact of the data in question had disappeared as well.

The practical evidence to support this assertion has been evident in the following recent cases:

- **FTC v. Everalbum:** The FTC concluded that the company that had been using the facial recognition technology without people's consent would be obliged to eliminate the software, not only the data themselves.
- **NYT v. OpenAI, Microsoft:** The allegations that GPT-4 preserved the copyrighted texts have revived the debate about whether the language models could delete the memory of private textual data after the completion of the machine learning procedure.
- **Clearview AI (2022):** European privacy enforcers-imposed penalties on Clearview AI Technologies Inc. over the use of unlawfully collected facial recognition data despite the company asserting that it had no reasonable means to verify that such data was deleted.

The reason behind these incidents is the same: AI systems do not support a native “delete” functionality and, even if approximate unlearning techniques are employed, there’s no way for the user or auditor to make sure the deletion occurred and the company made its proprietary model weights public.

CryptoForget represents a practical deep research and development project whose objective is to create a working and scaled-down version of the whole pipeline necessary for solving this problem

— including the domains of modern machine learning engineering, MLOps, distributed systems, applied cryptography, and privacy regulations and laws, using just one particular industry (FinTech credit risk/ loan default model will be illustrated in this paper).

Research Gap Analysis

Existing Research System	Limitation Identified	Research Gap	Proposed Solution in This Work
Traditional ML models	Once trained, user data remains embedded in model parameters.	No efficient mechanism to selectively remove an individual user's influence.	Implement SISA-based exact machine unlearning for selective data deletion.
GDPR/CCPA compliance tools	Deleting database records does not remove learned information from trained models.	Lack of verifiable machine learning data deletion.	Integrate machine unlearning with cryptographically verifiable deletion certificates.
Existing Machine Unlearning methods	Focus primarily on model retraining efficiency.	Limited emphasis on end-user transparency and auditability.	Generate signed deletion certificates backed by Merkle proofs and ECDSA signatures.
Cryptographic verification research	Primarily theoretical or focused on blockchain applications.	Limited integration with machine unlearning pipelines.	Combine Merkle commitments, audit logs, and machine unlearning into one framework.
MLOps platforms (MLflow, Feast, etc.)	Provide lifecycle management but lack privacy-aware retraining workflows.	No native support for right-to-be-forgotten requests.	Integrate deletion requests directly into the MLOps pipeline using FastAPI, Kafka, and Airflow/Prefect.
Existing privacy-preserving ML research	Focuses on federated learning, differential privacy, or secure inference.	Few end-to-end implementations combining MLOps, machine unlearning, and cryptographic verification.	Develop a complete end-to-end architecture integrating ML, MLOps, machine unlearning, cryptographic proofs, and user-facing dashboards.
Approximate unlearning techniques	Faster but may not guarantee complete removal of user influence.	Need practical comparison between exact and approximate unlearning.	Benchmark SISA exact retraining against Fisher-information scrubbing as a stretch goal.
Academic prototypes	Often evaluated only on algorithmic performance.	Limited focus on usability and user trust.	Provide React dashboards for users, administrators, and auditors with real-time deletion status and verification certificates.

Machine learning benchmark studies	Primarily evaluate model accuracy.	Lack of evaluation metrics for privacy and deletion effectiveness.	Evaluate MIA success rate, retraining cost, certificate generation time, deletion latency, and model utility retention.
Existing literature	Individual components are studied independently.	No unified architecture combining MLOps, cryptographic verification, machine unlearning, and regulatory compliance.	Propose a unified, modular, and deployment-ready architecture suitable for FinTech, Healthcare, and EdTech applications.

Problem Statement

How can a machine learning production system enable any user to delete their personal information, both from the ongoing education and from the model that has already received training in order to provide every user and auditor with cryptographic evidence of what has been deleted, while at the same time protecting the proprietary parameters of the model and anybody else's personal data?

It breaks down into four concrete sub-problems that the project should resolve:

- 1. Deletion in a working system rather than just experimental evaluations.** Generally speaking, most academic papers about unlearning work with persistent datasets (such as CIFAR, MNIST). Here, the situation is that a request for deletion must be implemented in a working system with users present, a live model registry, and ongoing retraining/drift cycles.
- 2. Two types of deletion.** (a) data not yet used for the current version of the model ("pre-training deletion" is skipping data collection before the next retraining), and (b) data already involved in the current model ("post-training deletion" which implies a real action of deletion/unlearning).
- 3. The need to verify the deletion without revealing any details.** The user should receive evidence that his data is no longer usable without needing CryptoForget to disclose all the details of the model and the data of the other users.
- 4. Evidence of deletion rather than a black-box verification.** The system must provide the user (and possibly some auditor) with an ability to see how much of the influence was removed without just saying "done".

Proposed Solution

CryptoForget combines three layers that are usually studied in isolation i.e. **MLOps engineering**, **machine unlearning algorithms**, and **applied cryptography** into one demonstrable pipeline.

3.1 The Two Deletion Paths

Path A is defined as the straightforward approach of pre-training removal. If the data of the user has not yet been included in the model version being currently deployed, then removal is straightforward and involves setting the respective records to tombstone in the feature store and excluding these records during the next retraining job.

Path B - Post-training removal (the difficult and fascinating scenario).

If the current model was trained using user data, the one proper way to resolve the problem is carrying out the whole process of training again, which is difficult and costly. The idea of CryptoForget relies on the method of un-learning. SISA (Sharded, Isolated, Sliced, Aggregated training) consists of splitting the training samples into sub-samples, so that each of these snapshots trains its particular model, and the outputs of the models are combined together. This minimizes the expenses when solving the problem using SISA's approach; in this case, only the sub-sample where the deletion occurred needs to be re-trained, which saves a lot of money in comparison to the whole model re-training process.

3.2 The Proposed System Design Flow



3.3 The Cryptographic Verification Layer

To make deletion *provable* rather than just *claimed*:

- 1. Commitment during ingestion:** every batch is converted to a hash and organized into a Merkle tree. The root hash is recorded in an append-only hash-chained audit log (which functions like a lightweight blockchain) prior to training.
- 2. Proof of inclusion (during consent):** a proof that validates the presence of the user's data in the model can be provided to the user in case the model is trained.
- 3. Proof of exclusion (after unlearning):** when a shard is retrained/scrubbed, the new root hash is calculated. CryptoForget creates a signed deletion certificate (ECDSA signature) containing: the old root, the new root, Merkle proof of the absence of the user's leaf in the tree, timestamp, and a hash of the new model registered in MLflow. This certificate can be independently verified by the user or an auditor without revealing the model weights or information of other users.
- 4. Empirical evidence rather than a black box:** the dashboard, in addition to the certificate, informs the user about concrete numbers before and after the procedure, such as the decreasing rate of success of membership inference attacks related to their data (ideally reaching around 50% in other words, random guessing), and accuracy of the model retained on the other part of the population so that it has an actual and measurable answer to the question "how much of my data has been removed" as opposed to just a promise often found in marketing slogans.

Note on cryptography: while having a zk-SNARK circuit that demonstrates arbitrary unlearning is a valid and interesting idea, this solution contains a heavy, PhD-level undertaking. This proposal involves Merkle commitments and digital signatures as an achievable cryptographic layer for this project and considers zk-SNARK for the purpose of verifiable unlearning for teams wishing to go far.

Implementation Plan

Phase	Duration	Deliverables
0. Setup & Domain Selection	Week 1–2	Finalize domain (FinTech loan-default / Healthcare readmission / EdTech dropout-risk); design database schema; set up AWS project; configure CockroachDB; create repository structure.
1. Core ML + MLOps Pipeline	Week 3–6	Develop baseline LightGBM (or sharded-ensemble) model; integrate MLflow tracking and model registry; configure Feast feature store; implement Evidently AI-based drift detection; create Airflow/Prefect retraining workflow (DAG).
2. User-Agent Simulation Layer	Week 5–7	Develop 15–20 synthetic/real user agents using Python and Faker; build FastAPI endpoints for onboarding, model inference, and profile management; integrate Kafka event bus for user activities.
3. Unlearning Engine	Week 7–10	Implement pre-training tombstoning; develop SISA-style sharded retraining; implement Fisher-scrubbing (stretch

		goal); build influence-quantification module using Membership Inference Attack (MIA) test harness.
4. Cryptographic Verification Layer	Week 9–12	Develop Merkle tree commitment library; implement hash-chained audit ledger; build ECDSA signing service; create deletion-certificate generator and verifier (CLI/API).
5. Dashboard & UX	Week 11–13	Develop React dashboard with user profile; implement "Stop/Delete My Data" functionality; integrate live deletion certificate viewer; build administrator/auditor dashboard with drift charts and audit trail.
6. Benchmarking & Evaluation	Week 13–15	Execute benchmark experiments; collect evaluation metrics; perform SISA shard-count vs. retraining-cost ablation study; analyze and document results.
7. Paper Writing & Submission	Week 15–16	Prepare IEEE/ACM-style research paper; finalize figures and experimental results; submit to conference/journal or department showcase.

Benchmarking Matrix

Metric	Target
Unlearning Completeness (MIA success rate on deleted user after scrubbing)	Converges toward ~ 50% (equivalent to random guessing)
Model Utility Retention (on remaining users)	< 2% drop in Accuracy/AUC compared to the pre-deletion model
Retraining Cost vs. Full Retraining (SISA shard-only retraining)	≥ 3–5× faster than full model retraining
Certificate Verification Time	< 100 ms
Certificate Size	A few KB (contains only the Merkle proof and ECDSA signature, not the model weights)
End-to-End Deletion Latency (request to certificate issuance)	Measured, reported, and discussed rather than constrained to a fixed target

Project Feasibility

As each of the parts is implemented through established and well-documented open-source technologies, the project is doable within a semester-size timeline.

- **SISA-style Exact Unlearning:** SISA (Sharded, Isolated, Sliced, and Aggregated) unlearning is a well-known technology first described in 2021 and with public implementations already available. The implementation of shard-based retraining for either a small neural network or a LightGBM ensemble is thus more an engineering task than research effort.
- **Cryptographic Verification:** Merkle trees and ECDSA digital signatures are widely adopted cryptographic primitives supported by mature Python libraries such as hashlib, cryptography, and ecdsa. The project does not require designing custom cryptographic protocols or zero-knowledge circuits, significantly reducing implementation complexity.
- **MLOps Infrastructure:** The MLOps stack that has been demonstrated would utilize some components being used in the team's current Retail Operations Intelligence project, thus allowing for deployment principles, infrastructure and experience from the prior development to be applied in this case.

- **Clearly Defined Scope:** Willingly limited scope: The main research-heavy part of the project, i.e., approximate machine unlearning using Fisher Scrubbing, has been specially chosen as a stretch goal instead of making it a must-have output. Therefore, the core goals are simply dependent on pre-training tombstoning combined with SISA-type exact unlearning and cryptographically secure deletion certificates.

Technologies Stack

Layer	Technology / Framework
ML Model	LightGBM / XGBoost (for tabular data); optionally a small sharded neural network for the deep learning variant
MLOps	MLflow (experiment tracking & model registry), Feast (feature store), Evidently AI (data/model drift detection), Airflow or Prefect (workflow orchestration)
Backend / API	FastAPI (Python) for REST APIs; Kafka for event streaming of user actions and deletion requests
Database	CockroachDB for user profiles, consent management, metadata, and data lineage
Unlearning Engine	Custom SISA-based sharding and retraining (LightGBM/PyTorch); Fisher Information Scrubbing (stretch goal)
Cryptographic Verification	Python hashlib and cryptography libraries for Merkle tree construction, ECDSA digital signatures, and a hash-chained append-only audit log
Frontend	React-based user portal and administrator/auditor dashboard
Cloud & DevOps	AWS, Docker, and GitHub Actions for CI/CD
Evaluation & Benchmarking	Membership Inference Attack (MIA) test harness, custom benchmarking scripts, and performance evaluation metrics

Conclusion

CryptoForget takes what is normally regarded as a vague legal principle, particularly the concept of “the right to be forgotten,” and comes up with a practical and verifiable technical implementation of it. By utilizing a real MLOps pipeline (including training, registry, drift detection, and automated retraining) combined with a community of simulated agents that create and use their personal data and an exact-unlearning engine executed using SISA-style sharding techniques and a compact yet real cryptographic verification measure (Merkle commitments and deletion certificates), the project produces that seldom found in the field of student research – a system a user can really click on and get a mathematically verifiable confirmation that his/her data influence is gone. The end result is identifiably possible to deliver on an academic schedule, utilizes the previously established patterns in technological infrastructure, and provides a clear approach (using zk-SNARK to implement unlearning) for others who would want to develop it further as a thesis project.

References

1. Foundational Machine Unlearning

[1] Bourtole, L., Chandrasekaran, V., Choquette-Choo, C. A., et al. (2021). *Machine Unlearning*. Proceedings of the IEEE Symposium on Security and Privacy. Introduces the SISA

(Sharded, Isolated, Sliced, and Aggregated) training methodology for efficient exact machine unlearning. Available: <https://arxiv.org/abs/1912.03817>

[2] Guo, C., Goldstein, T., Hannun, A., & van der Maaten, L. (2020). *Certified Data Removal from Machine Learning Models*. Proceedings of the International Conference on Machine Learning (ICML). Available: <https://arxiv.org/abs/1911.03030>

[3] Mercuri, S., Khraishi, R., Okhrati, R., et al. (2022). *An Introduction to Machine Unlearning*. Available: <https://arxiv.org/abs/2209.00939>

2. Survey Papers

[4] Nguyen, T. T., et al. (2022). *A Survey of Machine Unlearning*. Available: <https://arxiv.org/abs/2209.02299>

[5] Qu, Y., Yuan, X., Ding, M., et al. (2023). *Learn to Unlearn: A Survey on Machine Unlearning*. Available: <https://arxiv.org/abs/2305.07512>

[6] Shaik, T., Tao, X., Xie, H., et al. (2023). *Exploring the Landscape of Machine Unlearning: A Comprehensive Survey and Taxonomy*. Available: <https://arxiv.org/abs/2305.06360>

[7] Xu, H., Zhu, T., Zhang, L., et al. (2023). *Machine Unlearning: A Survey*. Available: <https://arxiv.org/abs/2306.03558>

[8] Wang, W., et al. (2024). *Machine Unlearning: A Comprehensive Survey*. Available: <https://arxiv.org/abs/2405.07406>

3. Approximate and Certified Machine Unlearning

[9] Golatkar, A., Achille, A., & Soatto, S. (2020). *Eternal Sunshine of the Spotless Net: Selective Forgetting in Deep Networks*. Proceedings of CVPR 2020.

[10] Kurmanji, M., Triantafillou, P., Hayes, J., & Triantafillou, E. *Towards Unbounded Machine Unlearning*. Available: <https://openreview.net/pdf?id=OveBaTtUAT>

4. Zero-Knowledge and Cryptographic Verification

[11] Kang, D., et al. (2022). *Scaling up Trustless DNN Inference with Zero-Knowledge Proofs*. Proceedings of the ACM Conference on Computer and Communications Security (CCS).

5. Technical Reports

[12] Liu, K. Z. (2024). *Machine Unlearning in 2024*. Stanford University Computer Science Technical Report. Available: <https://ai.stanford.edu/~kzliu/files/unlearning.pdf>

6. Regulatory and Legal References

[13] General Data Protection Regulation (GDPR), Regulation (EU) 2016/679, Article 17 – *Right to Erasure ("Right to be Forgotten")*.

[14] California Consumer Privacy Act (CCPA), Cal. Civ. Code §1798.105 – *Right to Delete*.

[15] Federal Trade Commission (FTC). *FTC v. Everalbum, Inc.* (2021). Enforcement action introducing the concept of algorithmic disgorgement.